

T3 Field Sheet — Hull-Only Ballast Validation

Print this sheet and bring it to the test. Background, rationale, and pass conditions are in `tests.md`.

Pre-Requisites

- ☒ Sealed hull (both lids, syringe hole blocked — record method below)
- ☒ Kitchen scale accurate to 1 g
- ☒ Lead shot and a small weighing pot/cup
- ☒ Water container large enough to fully submerge the hull (250 mm × 110 mm)
- ☒ Something to dry the hull between steps
- ☒ Phone or audio recorder for observations (or write in the Observations section)

Syringe hole blocking method: BLU TACK

Step 0 — Weigh the Empty Hull

Weigh the assembled, sealed, empty hull before adding any ballast.

Measurement	Value
Empty hull mass	<u>455</u> g
Predicted ballast (= 2376 – hull mass)	<u>1921</u> g
Phase 1 start (= predicted – 500 g)	<u>1421</u> g

(some blowed 45g)

Phase 1 — Coarse (250 g steps)

Start at the Phase 1 start value from Step 0. Add 250 g at each step. Stop when the hull sinks or is clearly at the transition.

Float state key: **F** = floats clearly / **N** = near waterline / **S** = sinks

Tilt key: **L** = level / **B** = bow up / **St** = stern up / **R** = rolled

Step	Total ballast (g)	Float (F/N/S)	Tilt (L/B/St/R)
1.1	1421	F	L (slight tilt) (sits quite low)
1.2	1671	F	L (low)
1.3	1921	N	L (v. low)
1.4	2171		
1.5			
1.6			

Phase 1 bracket: floats at _____ g, sinks at _____ g

Phase 2 — Medium (50 g steps)

Start at the last clearly-floating mass from Phase 1. Add 50 g at a time.

Step	Total ballast (g)	Float (F/N/S)	Tilt (L/B/St/R)
2.1	1971	F	L
2.2	2021	S	L
2.3			
2.4			
2.5			
2.6			

Phase 2 bracket: floats at _____ g, sinks at _____ g

Phase 3 — Fine (10 g steps)

Start at the last clearly-floating mass from Phase 2. Add 10 g at a time.

Step	Total ballast (g)	Float (F/N/S)	Tilt (L/B/St/R)
3.1	1981	F	L
3.2	1991	F	L
3.3	2001	F	L
3.4	2011	S	L
3.5			
3.6			

(tiny bit off the bottom)

Phase 3 bracket: floats at _____ g, sinks at _____ g

Phase 4 — Precision (1 g steps)

Start at the last clearly-floating mass from Phase 3. Add 1 g at a time.

Step	Total ballast (g)	Float (F/N/S)	Tilt (L/B/St/R)
4.1	2002	S N	L
4.2	2003	N	L
4.3	2004	N	L
4.4	2005	S	L
4.5			
4.6			
4.7			
4.8			
4.9			
4.10			

Result

	Value
Last mass that clearly floats	<u>2004</u> g
First mass that clearly sinks	<u>2005</u> g
Empty hull mass (from Step 0)	<u>455</u> g
Implied displaced volume (= sink mass / 1.0)	<u>2460</u> cm ³
Predicted displaced volume (from calc)	2376 cm ³
Difference	<u>84</u> cm ³

Observations

(Reference steps by number, e.g. 2.3. Leave blank if using audio recording.)

1.2. Water clearly entering around valve.

Water = $d = 33\text{ cm}$
 $l = 41\text{ cm}$
 $w = 59\text{ cm}$

~~40~~ Accuracy of the scales is questionable, sometimes skipping 2g at a time.

Final mass, however, rechecked at 2005g.